

PHU LIEN, Viet Nam

WMO: 488260

Lat: 20.795N

Lon: 106.614E

Elev: 116

StdP: 99.94

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.8	10.9	4.8	5.4	13.5	6.5	6.1	13.5	5.3	17.2	4.8	16.4	1.9	50	0.288

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	5.4	34.3	28.9	33.3	28.7	32.5	28.4	30.0	32.8	29.4	32.2	28.8	31.5	2.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
29.2	26.4	32.1	28.7	25.5	31.4	28.1	24.7	30.6	101.2	32.9	97.9	32.3	95.2	31.7	32.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.2	5.2	4.4	DB	7.8	36.7	2.7	1.6	5.9	37.8	4.3	38.7	2.7	39.6	0.8	40.7
			WB	6.7	30.6	1.4	0.8	5.6	31.2	4.8	31.7	4.0	32.2	3.0	32.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.0	16.9	18.0	20.2	24.0	27.2	29.1	28.8	28.3	27.5	25.8	22.6	18.7
	DBStd	5.01	3.40	3.59	2.96	2.58	2.17	1.75	1.74	1.71	1.70	2.09	2.83	3.04
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	169	68	47	16	1	0	0	0	0	0	0	4	32
	CDD10.0	5098	215	224	317	421	534	572	584	568	525	491	377	270
	CDD18.3	2224	24	38	75	172	276	322	326	309	275	233	131	44
	CDH23.3	20302	21	51	131	939	2687	3883	3869	3438	2771	1809	614	89
	CDH26.7	6917	0	3	8	142	858	1638	1563	1247	893	467	93	3
Wind	WSAvg	2.0	1.7	1.7	1.8	2.1	2.3	2.3	2.3	2.1	2.2	2.1	2.1	1.9
Precipitation	PrecAvg	1621	27	21	49	69	197	242	279	323	249	93	47	24
	PrecMax	2202	168	70	174	194	467	435	599	629	404	612	222	81
	PrecMin	963	0	1	0	18	77	143	91	123	66	0	0	0
	PrecStd	280	37	14	39	47	94	72	117	143	90	123	46	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.3	26.6	27.7	31.5	35.2	35.7	35.4	34.6	33.2	32.0	30.1	27.1
		MCWB	22.2	23.5	24.6	27.9	29.1	28.6	29.0	29.2	28.6	27.0	25.6	23.9
	2%	DB	23.7	24.8	26.0	29.6	33.0	34.3	33.8	33.4	32.4	31.0	28.8	25.6
		MCWB	20.9	22.3	23.7	26.5	28.6	28.6	28.9	28.9	28.2	26.3	24.4	22.6
	5%	DB	22.5	23.6	24.7	28.3	31.6	33.1	32.8	32.4	31.6	30.2	27.7	24.2
		MCWB	20.2	21.6	22.9	25.8	27.9	28.7	28.7	28.5	27.7	25.7	23.7	21.4
	10%	DB	21.4	22.5	23.7	27.2	30.4	32.0	31.8	31.4	30.7	29.3	26.6	23.0
		MCWB	19.6	21.2	22.5	25.1	27.2	28.3	28.2	28.1	27.1	25.1	23.1	20.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.1	24.2	25.3	28.2	30.1	30.6	30.7	30.5	29.9	28.3	26.9	24.9
		MCDB	24.5	26.1	26.7	30.7	33.3	33.3	33.4	33.2	32.4	30.8	29.2	26.4
	2%	WB	21.8	22.9	24.2	27.1	29.1	29.7	29.7	29.6	29.0	27.4	25.7	23.6
		MCDB	23.2	24.1	25.3	29.1	32.2	32.5	32.6	32.3	31.6	30.0	27.5	24.9
	5%	WB	20.9	22.2	23.4	26.2	28.3	29.1	29.1	28.9	28.2	26.6	24.9	22.3
		MCDB	22.0	23.1	24.4	27.8	30.9	31.8	31.9	31.5	30.7	29.1	26.6	23.5
	10%	WB	20.0	21.4	22.8	25.5	27.6	28.5	28.4	28.3	27.6	25.9	24.2	21.1
		MCDB	21.0	22.3	23.5	26.8	29.8	31.1	31.0	30.7	29.9	28.1	25.8	22.2
Mean Daily Temperature Range	5% DB	MDBR	4.8	4.5	4.1	4.6	5.5	5.4	5.1	5.2	5.7	6.2	6.2	5.7
		MCDBR	6.6	5.9	5.5	5.9	7.0	6.6	6.5	6.7	6.8	7.2	7.3	6.9
		MCWBR	4.1	3.6	3.5	3.5	3.6	3.0	3.1	3.3	3.5	3.4	3.8	4.1
	5% WB	MCDBR	5.8	4.9	4.6	5.3	6.4	6.0	6.2	6.3	6.3	6.5	6.2	5.8
		MCWBR	4.1	3.4	3.4	3.4	3.7	3.3	3.5	3.7	3.7	3.6	3.8	4.1
Clear-Sky Solar Irradiance	taub	0.589	0.625	0.789	0.743	0.547	0.509	0.483	0.536	0.584	0.681	0.584	0.612	
	taud	1.772	1.737	1.476	1.592	2.083	2.210	2.277	2.099	1.933	1.680	1.840	1.750	
	Ebn at Noon	680	688	600	640	772	795	817	778	735	637	680	642	
	Edh at Noon	207	226	304	274	166	145	136	163	190	236	192	205	
All-Sky Solar Radiation	RadAvg	2.29	2.62	2.86	3.79	4.85	5.04	4.95	4.58	4.41	3.98	3.35	2.61	
	RadStd	0.56	0.44	0.36	0.41	0.42	0.42	0.51	0.35	0.30	0.25	0.43	0.42	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.41	+0.35	+0.40	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	+0.56	N/A	N/A	N/A	N/A	N/A	+80

Nomenclature: See separate page